RAHUL ARORA

Research Scientist — Computer Graphics & Human-Computer Interaction

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EXECUTIVE SUMMARY

Research Scientist experienced in architecting multimodal AI wearables and end-to-end 3D graphics systems. Expert in owning sensing stacks (firmware to front-end) and driving ML research through ground-truth algorithms and user experiments. Widely published and patented in computer graphics, AR/VR, and HCI.

CORE COMPETENCIES & TECHNICAL STACK

- **Domains:** HCI, Computer Graphics, AR/VR, Sensing, Gestural Interfaces.
- **Programming Languages:** Unity/C#, Python, C++, TypeScript/React, MATLAB.
- **Techniques:** Sensor Fusion, Spatial Tracking, Inertial Measurement Units (IMUs), Numerical Optimization, Geometry Processing, Quantitative/Qualitative User Studies, Rapid Prototyping.

PROFESSIONAL EXPERIENCE

∞ META REALITY LABS — Dec 2021 – Jul 2026

Research Scientist (Input Explorations Team)

- **Meta Neural Band:** Led R&D for continuous-control gestures on Meta Ray-Ban Display glasses, from sensor fusion and gesture tracking through user studies, embedded implementation, and OS/application integration. Helped bring the interaction to product release and contributed to the related Nature (2025) neuromotor-interface research.
- **Multimodal Interaction:** Designed and evaluated device-control and text-input concepts combining surface electromyography, gaze, head tracking, and computer vision. Contributed to IMU and computer-vision fused hand gestures for Orion AR glasses and to eye-tracking-driven interactions for Meta VR Glasses. Studied hand–eye coordination and gaze fixation to understand and mitigate selection errors in AR/VR.
- **3D Math and Head Gestures:** Served as a primary resource for rotations and spatial reasoning across Wearables, a multi-thousand strong business unit. Resolved on-device orientation-estimator issues, developed IMU-based head-gesture models, improved gesture reliability via geometric analysis, and aligned coordinate conventions across teams.
- **Ground-Truth Systems:** Directed wearable firmware and custom data-collection tools for neuromotor research. Built an IMU-based alternative to a motion-capture rig that increased data throughput by 2–3× while reducing collection cost; designed human-subject studies to measure sensing performance.
- **Evaluation Tools:** Built reusable analysis pipelines, telemetry, visualizations, and developer-facing tools to inspect real-world failures and measure interaction reliability across hundreds of thousands of interactions.
- **Engineering Quality:** Added tests using real sensor data, standardized geometric units, math libraries, and coordinate systems, and removed more than 10,000 lines of legacy code. Reduced broken tests by 92% and accelerated recurring notebook analyses by 20–50× through caching and parallelization.

ADOBE RESEARCH — Summers 2019, 2018, 2013

Research Intern (Multiple Teams)

- **2019 — Stylized Fluid Animation:** Developed interpretation and interpolation algorithms that turn a few keyframes into stylized 2D fluid animations.
- **2018 — Gestural Animation in VR:** Elicited gestures from professional animators and built a VR animation system combining gestural performance with keyframing. Published as *MagicalHands* at UIST 2019.

- **2013 — Computer Vision Data:** Built and publicly deployed a gamified crowdsourcing platform to accelerate and improve object-recognition tasks.

AUTODESK RESEARCH **Winter 2017 & Summer 2016**
Research Intern (User Interface Research Team)

- **2017 — SymbiosisSketch:** Developed an AR concept-sketching system that used motion capture to map a physical tablet to virtual surfaces for detailed 3D drawing. Evaluated it with professional artists; published at CHI 2018 and was granted a US patent.
- **2016 — Sketching Perception:** Designed lab experiments and quantitative analyses to establish how human perception and motor control affect the accuracy of 3D sketching in VR. Published at CHI 2017.

JanusVR **Winter 2016**
Independent Contractor

- **VR Scripting Runtime:** Extended JanusVR's JavaScript engine for domain-specific, DOM-like scene objects: shipped hierarchy traversal and object lookup APIs, fixed various memory leaks, crashes, and serialization bugs.

EDUCATION

UNIVERSITY OF TORONTO **Sep 2015 – Jul 2021**
Ph.D. in Computer Science; Advisor: Prof. Karan Singh, Dynamic Graphics Project Lab

- **Focus:** Computer Graphics and HCI.
- **Thesis:** *Creative Visual Expression in Immersive 3D Environments* — Guidelines, algorithms, and tools for overcoming the perceptual and musculoskeletal limits of mid-air input for AR/VR creative workflows.

INDIAN INSTITUTE OF TECHNOLOGY (IIT) KANPUR **Jul 2010 – Jul 2015**
Bachelor and Masters of Technology (B.Tech. & M.Tech.) in Computer Science and Engineering

- **Masters Thesis:** *Exploring Design Space by Interpolating Between Multiple Sketches* — Developed a novel technique for interpolating between multiple early-stage design concepts.
- **Undergraduate Research:** Improved the computational complexity of the perfect matching problem for $K_{3,3}$-free or K_5-free graphs, an important class of non-planar graphs.

PEER-REVIEWED PUBLICATIONS

A Generic Noninvasive Neuromotor Interface for Human-Computer Interaction
CTRL-Labs at Reality Labs (245 contributors including **Rahul Arora**)
Nature, 2025
https://www.nature.com/articles/s41586-025-09255-w

Piecewise-Smooth Surface Fitting onto Unstructured 3D Sketches
Emilie Yu, **Rahul Arora**, J. Andreas Bærentzen, Karan Singh, and Adrien Bousseau
ACM Transactions on Graphics (TOG) 2022, proc. SIGGRAPH
https://em-yu.github.io/research/surfacing_3d_sketches/

Mid-Air Drawing of Curves on 3D Surfaces in Virtual Reality
Rahul Arora and Karan Singh
ACM Transactions on Graphics (TOG) 2021, presented at SIGGRAPH 2021
http://bit.ly/tog21_mimicry

CASSIE: Curve and Surface Sketching in Immersive Environments
Emilie Yu, **Rahul Arora**, Tibor Stanko, J. Andreas Bærentzen, Karan Singh, and Adrien Bousseau
ACM SIGCHI Conference on Human Factors in Computing Systems 2021 (CHI '21)
Best Paper Honorable Mention
https://em-yu.github.io/research/cassie/

MagicalHands: Mid-Air Hand Gestures for Animating in VR
Rahul Arora, Rubaiat Habib Kazi, Danny Kaufman, Wilmot Li, and Karan Singh
ACM Symposium on User Interface Software and Technology 2019 (UIST '19)
https://www.dgp.toronto.edu/projects/magical-hands/

Volumetric Michell Trusses for Parametric Design & Fabrication
Rahul Arora, Alec Jacobson, Timothy R. Langlois, Yijiang Huang, Caitlin Mueller, Wojciech Matusik, Ariel Shamir, Karan Singh, and David I.W. Levin
ACM Symposium on Computational Fabrication 2019 (SCF '19)
https://www.dgp.toronto.edu/projects/michell/

SymbiosisSketch: Combining 2D and 3D Sketching for Designing Detailed 3D Objects in Situ
Rahul Arora, Rubaiat Habib Kazi, Tovi Grossman, George Fitzmaurice, and Karan Singh
ACM SIGCHI Conference on Human Factors in Computing Systems 2018 (CHI '18)
https://doi.org/10.1145/3173574.3173759

Experimental Evaluation of Sketching on Surfaces in VR
Rahul Arora, Rubaiat Habib Kazi, Fraser Anderson, Tovi Grossman, Karan Singh, and George Fitzmaurice
ACM SIGCHI Conference on Human Factors in Computing Systems 2017 (CHI '17)
http://dx.doi.org/10.1145/3025453.3025474

SketchSoup: Exploratory Ideation using Design Sketching
Rahul Arora, Ishan Darolia, Vinay P. Namboodiri, Karan Singh, and Adrien Bousseau
Computer Graphics Forum (CGF) 2017, presented at Eurographics 2017
http://dx.doi.org/10.1111/cgf.13081

Derandomizing Isolation Lemma for $\mathbf{K}_{3,3}$-free and $\mathbf{K}_5$-free Bipartite Graphs
Rahul Arora, Ashu Gupta, Rohit Gurjar, and Raghunath Tewari
Symposium on Theoretical Aspects of Computer Science (STACS) 2016
http://dx.doi.org/10.4230/LIPIcs.STACS.2016.10

OTHER PUBLICATIONS

Introduction to 3D Sketching (Invited Book Chapter)
Rahul Arora, Mayra Donaji Barrera Machuca, Philipp Wacker, Daniel Keefe, and Johann Habakuk Israel
In Interactive Sketch-Based Interfaces and Modelling for Design. River Publishers. 2023.
https://dx.doi.org/10.1201/9781003360650-8

Input Processing and Geometric Representations for 3D Sketches (Invited Book Chapter)
Johann Habakuk Israel, Mayra Donaji Barrera Machuca, **Rahul Arora**, Philipp Wacker, and Daniel Keefe
In Interactive Sketch-Based Interfaces and Modelling for Design. River Publishers. 2023.
https://dx.doi.org/10.1201/9781003360650-9

Interaction Devices and Techniques for 3D Sketching (Invited Book Chapter)
Mayra Donaji Barrera Machuca, **Rahul Arora**, Philipp Wacker, Daniel Keefe, and Johann Habakuk Israel
In Interactive Sketch-Based Interfaces and Modelling for Design. River Publishers. 2023.
https://dx.doi.org/10.1201/9781003360650-10

3D Sketching Application Scenarios (Invited Book Chapter)
Philipp Wacker, **Rahul Arora**, Mayra Donaji Barrera Machuca, Daniel Keefe, and Johann Habakuk Israel
In Interactive Sketch-Based Interfaces and Modelling for Design. River Publishers. 2023.
https://dx.doi.org/10.1201/9781003360650-11

Thinking Outside the Lab: VR Size & Depth Perception in the Wild (Preprint)
Rahul Arora, Jiannan Li, Gongyi Shi, Karan Singh
https://arxiv.org/2105.00584

Creative Expression with Immersive 3D Interactions (Juried)
Rahul Arora
Extended Abstracts of the 2020 CHI Conference (CHI '20 Doctoral Consortium)
https://dl.acm.org/doi/10.1145/3334480.3375028

Designing Volumetric Truss Structures for Computational Fabrication (Juried)
Rahul Arora, Alec Jacobson, Timothy R. Langlois, Karan Singh, and David I.W. Levin
Graphics Interface 2018 Posters (GI '18)
https://bit.ly/31CTjtw

PATENTS

Techniques for using in-air wrist-roll gestures detected via a wrist-wearable device to operate another device, and wearable devices and systems for performing those techniques
Christopher Anderson, **Rahul Arora**, and Julian Andres Ramos Rojas
US Patent 12,695,975; granted 2026
https://patents.google.com/patent/US12695975B2/en

Techniques for presenting augmented-reality content in multiple presentation modes based on real-world context, and devices, systems, and methods thereof
Hayden Schoen, Christopher Anderson, **Rahul Arora**, Nathan Aschenbach, Pol Pla, Luke O'Connor, Jenna Velez, Israel Pastrana Vicente, and Shengzhi Wu
US Patent Application 19/244,940; published 2025
https://patents.google.com/patent/US20250390198A1/en

Occlusion Avoidance of Virtual Objects in an Artificial Reality Environment
Chenxin Hong, Philipp Schoessler, Bruno de Araujo, **Rahul Arora**, Justin Ryan Reid, and Dan Kun-yi Chen
US Patent Application 18/906,940; published 2025
https://patents.google.com/patent/US20250200904A1/en

Computer-aided techniques for designing detailed three-dimensional objects
Karansher Singh, Tovi Grossman, Kazi Rubaiat Habib, George Fitzmaurice, and **Rahul Arora**
US Patent 11,074,747; granted 2021
https://patents.google.com/patent/US11074747B2/en

AWARDS AND RECOGNITION

CHI Best Paper Honorable Mention Award	2021
Robert C. Lansdale/Okino Computer Graphics Fellowship: CA $4,000	2020
Wolfond Scholarship in Wireless Information Technology: CA $10,000	2019
UofT Libraries Grad Exhibit Competition (3 winners): CA $1,000	2019
Adobe Research Fellowship 2019 (11 fellows): US $10,000	2018

Adobe Research Fellowship 2018 Finalist 2017

Mitacs Accelerate Award for industrial partnership: CA $15,000 2016

Merit-cum-Means Scholarship, IIT Kanpur: Full tuition amount 2010–2014

Academic Excellence Award, IIT Kanpur (top 5% students) 2013

TALKS

New York University HCI Course, Remote Talk November 2023
On Immersive 3D Sketching and Modelling

Université de Montréal HCI Course, Remote Talk April 2022
On Immersive 3D Sketching and Modelling

Toronto Geometry Colloquium, Remote Talk October 2021
On Human-Centered Graphics for Immersive Creative Expression

SIGGRAPH 2021, Remote Talk August 2021
Paper presentation: Mid-Air Drawing of Curves on 3D Surfaces in Virtual Reality

Autodesk Research, Remote Talk April 2021
On Human-Centered Graphics for Immersive Creative Expression

Facebook Reality Labs, Remote Talk April 2021
On Human-Centered Graphics for Immersive Creative Expression

UBC AR/VR Course, Remote Guest Lecture March 2021
On Gesture-Based Animation in VR

GraphDeco Group, Inria Sophia-Antipolis, Remote Talk March 2021
On Human-Centered Graphics for Immersive Art & Design

MIT Computer Graphics Group, Remote Talk December 2020
On Volumetric Michell Trusses

CHI 2020 Doctoral Symposium, Remote Talk May 2020
On Creative Expression with Immersive 3D Interactions

Motograph Workshop, Waterloo, Canada December 2019
On Stylized Fluid Animation

UIST 2019, New Orleans, USA October 2019
Paper presentation: MagicalHands

SCF 2019, Pittsburgh, USA June 2019
Paper presentation: Volumetric Michell Trusses for Parametric Design & Fabrication

CHI 2018, Montreal, Canada May 2018
Paper presentation: SymbiosisSketch

Toronto SIGCHI Chapter, Toronto, Canada March 2018
On Hybrid 2D–3D Sketching in SymbiosisSketch

Tomograph Workshop, Toronto, Canada December 2017
On Truss Topology Optimization for Design & Manufacturing

IIT Kanpur, India May 2017
On 2D Sketching and Immersive 3D Sketching

CHI 2017, Denver, USA May 2017
Paper presentation: Experimental Evaluation of Sketching on Surfaces in VR

Eurographics 2017, Lyon, France May 2017
Paper presentation: SketchSoup

SERVICE

Committee Member for CHI 2022 and 2023 Late-Breaking Work.

Committee Member for ICCV 2021 Workshop on Sketching for Human Expressivity.

Committee Member for Pacific Graphics 2021 and 2022.

Committee Member for SIGGRAPH Asia 2020 XR Program.

Area Chair for Graphics Interface (GI) 2020.

Student Volunteer at User Interface Software and Technology (UIST) 2019.

Reviewer for computer graphics conferences
SIGGRAPH 2020, 2022–2025; SIGGRAPH Asia 2020–2024; SIGGRAPH Posters Program 2021; Eurographics (EG) Short Papers 2021; Symposium for Computational Fabrication (SCF) 2020; Pacific Graphics 2021–2022; Graphics Interface (GI) 2018, 2020.

Reviewer for computer graphics journals
Transactions on Graphics (TOG) 2021; Computer Graphics Forum (CGF) 2020, 2023–2024; Transactions on Visualization and Computer Graphics (TVCG) 2019–2020, 2023, 2025; Computer-Aided Design (CAD) 2019; Computer & Graphics 2019; Computer Graphics & Applications 2017.

Reviewer for HCI conferences
Conference on Human Factors in Computing Systems (CHI) 2017–2025; User Interfaces Software and Technology (UIST) 2018–2020, 2022–2024; Designing Interactive Systems (DIS) 2018; Creativity & Cognition 2022; Graphics Interface (GI) 2020.

Reviewer for HCI journals
International Journal of Human-Computer Interaction (IJHCI) 2018; International Journal of Human-Computer Studies (IJHCS) 2021.

Reviewer for specialized VR/AR conferences
Virtual Reality (IEEE VR) 2018, 2020–2023; Virtual Reality Software and Technology (VRST) 2020; International Symposium on Mixed and Augmented Reality (ISMAR) 2020–2022; Spatial User Interaction (SUI) 2017.

Reviewer for other venues
Springer Nature Applied Sciences (SNAS) 2019.

TEACHING

Computer Graphics, University of Toronto Winter 2020
Teaching Assistant for Prof. David Levin

Computer Graphics, University of Toronto Fall 2019
Teaching Assistant for Prof. Alec Jacobson

Computer Graphics, University of Toronto Winter 2019
Teaching Assistant for Prof. David Levin

Computer Graphics, University of Toronto Winter 2018
Teaching Assistant for Prof. Karan Singh and Prof. David Levin

Computer Graphics, University of Toronto — Fall 2017
Teaching Assistant for Prof. Karan Singh and Prof. Alec Jacobson

Intro to Theory of Computation, University of Toronto — Fall 2015
Teaching Assistant for Prof. Azadeh Farzan

Introduction to Computer Graphics, IIT Kanpur — Fall 2014
Teaching Assistant for Prof. Vinay P. Namboodiri

REFERENCES

Karan Singh
Professor of Computer Science, University of Toronto
PhD advisor
karan@dgp.toronto.edu

Rubaiat Habib Kazi
Sr. Research Scientist, Adobe
Internship mentor: Autodesk ('16, '17), Adobe ('18, '19)
rubaiat.habib@gmail.com

Christopher Collins
Research Scientist Manager, Meta
Manager at Meta (2025–2026)
christopher.m.collins@gmail.com

Jason Reisman
Principal Researcher, Fauna Robotics
Manager at Meta (2021–2024)
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Peter Walkington
Staff Research Engineer, Meta
Teammate at Meta (2021–2026)
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